

Section 1: Multiple Choice Questions

1. Which of the following is a property of the electromagnetic force?
 - a. Acts only over short distances
 - b. Responsible for binding protons and neutrons in the nucleus
 - c. Involves charged particles and affects atoms and molecules
 - d. Increases as distance increases
 2. An electron has a relative charge of:
 - a. +1
 - b. 0
 - c. -1
 - d. Undefined
 3. The gravitational force between two objects increases if:
 - a. Their masses decrease
 - b. The distance between them increases
 - c. Their masses increase
 - d. They are in a vacuum
 4. An isotope of carbon has a mass number of 14. How many neutrons does it have?
 - a. 6
 - b. 7
 - c. 8
 - d. 14
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Forces

5. Which force is responsible for keeping electrons in orbit around the nucleus?
 - a. Gravitational force
 - b. Electromagnetic force
 - c. Weak nuclear force
 - d. Strong nuclear force
6. If the distance between two objects is halved, the gravitational force will:
 - a. increase
 - b. decrease
7. If the mass of an object is doubled, the gravitational force will:
 - a. increase
 - b. decrease

Periodic Table and Elements

6. Which of the following elements is a noble gas?
 - a. Oxygen
 - b. Neon
 - c. Sodium
 - d. Hydrogen
 7. What information does the atomic number of an element provide?
 - a. The total number of protons and neutrons in the nucleus
 - b. The number of protons in the nucleus
 - c. The mass of the atom
 - d. The number of electrons in the atom
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Calculations and Problem Solving

8. A person has a mass of 70 kg. What is their weight on Earth? (Use $g=9.8$)
 - a. 686 N
 - b. 70 N
 - c. 7 N
 - d. 700 N
 9. How many moles are in 50 g of water (H_2O molar mass = 18 g/mol)?
 - a. 1.8
 - b. 2.7
 - c. 0.9
 - d. 5.0
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Section 2: Short Answer Questions

4. Force and Scale:

- o Name the force that holds planets in orbit around the Sun and explain why this force is weaker at greater distances.

5. Atomic Structure:

- o Describe what happens to an atom's charge, weight, and identity if it gains one proton.

6. Mass vs. Weight:

- o Imagine you are on Mars, where $g=3.7\text{N/kg}$, if your mass is 50 kg:
 - a. Calculate your weight on Mars.

b. Explain why your weight differs on Mars compared to Earth.

c. Explain why your weight is a tiny bit lower when standing on top of mount Everest on the earth, in comparison to the ground floor in Holland.

Section 3: Atomic Structure and the Periodic Table

7. There is an atom with an atomic number of 12, a mass number of 25.
 - a. How many protons in this atom:
 - b. How many neutrons in this atom:
 - c. How many electrons in this atom:
 - d. What atom is it:
8. Draw the electron shell diagram for a calcium atom.

Section 4: Analytical Thinking

9. A scientist claims that gravitational force is the weakest of the four fundamental is this true? And which one is the strongest?
10. A Uranium rod weighs 1837 g.
 - o Using the molar mass of Uranium calculate the number of moles of Uranium in the rod.
 - o How many Uranium atoms are in the rod? (Use 6.02×10^{23})

Section 5: Data Interpretation

11. Draw a graph which shows the relationship between the distance from the Earth's center and gravitational field strength:
- o Explain why this trend occurs using Newton's Law of Gravitation. (F_g)
12. A moon is orbiting a planet at a distance of 500,000 km. If the gravitational force between the moon and the planet is 2.5×10^{20} N, what happens to the force if the distance doubles? Show your reasoning.